

**A WEB APPLICATION
HUMAN RESOURCE MANAGEMENT SYSTEM**

By

Kishan Dervaliya (22CEUBG074)

&

Dev Patel (22CEUOS151)

A project submitted

In

**Partial fulfillment of the requirements
for the degree of**

**BACHELOR OF TECHNOLOGY
in
Computer Engineering**

Internal Guide

Prof. Ami M. Shah
Assistant Professor
Department of Computer Engineering

External Guide

Mr. Yash Shah
Team Lead(WMS)
Roima Intelligence



**Faculty of Technology
Department of Computer Engineering
Dharmsinh Desai University
April 2026**

CERTIFICATE

This is to certify that the project work titled
HUMAN RESOURCE MANAGEMENT SYSTEM

is the Bonafide work of

KISHAN B. DERVALIYA (22CEUBG074)

Carried out in the partial fulfillment of the degree of Bachelor of Technology in Computer Engineering at Dharmsinh Desai University in the academic session December 2025 to April 2026.

Prof. Ami M. Shah

Assistant Professor
Dept. of Computer Engineering

Dr. C. K. Bhensdadia

Head
Dept. of Computer Engineering



**Faculty of Technology
Department of Computer Engineering
Dharm Singh Desai University
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DEV D. PATEL (22CEUOS151)

Carried out in the partial fulfillment of the degree of Bachelor of Technology in Computer Engineering at Dharmsinh Desai University in the academic session December 2025 to April 2026.

Prof. Ami M. Shah

Assistant Professor
Dept. of Computer Engineering

Dr. C. K. Bhensdadia

Head
Dept. of Computer Engineering



**Faculty of Technology
Department of Computer Engineering
Dharmsinh Desai University
April 2026**

COMPANY CERTIFICATE

ROIMA

Date: 25.03.2025

TO WHOMSOEVER IT MAY CONCERN

Project Completion Certificate

This is to certify that Mr. Kishan Dervaliya, a student of Computer Engineering at DDU college, has successfully completed a project "Human Resource Management System" at Roima Intelligence India Pvt Ltd from 19th January 2026 to 21st March 2026. His internship with the organisation is currently ongoing. During his internship, he has worked with technologies like .NET, C#, React, SQL Server, Java and Spring Boot.

During this period, he has shown a good amount of responsibility, sincerity, and a genuine willingness to learn. He has also shown zeal to take on various assignments & challenges.

For Roima Intelligence India Pvt Ltd.


Disha Thakkar
HR Department





COMPANY CERTIFICATE

ROIIMA

Date: 25.03.2025

TO WHOMSOEVER IT MAY CONCERN

Project Completion Certificate

This is to certify that Mr. Dev Patel, a student of Computer Engineering at DDU college, has successfully completed a project "Human Resource Management System" at Roima Intelligence India Pvt Ltd from 19th January 2026 to 21st March 2026. His internship with the organisation is currently ongoing. During his internship, he has worked with technologies like .NET, C#, React, SQL Server, Java and Spring Boot.

During this period, he has shown a good amount of responsibility, sincerity, and a genuine willingness to learn. He has also shown zeal to take on various assignments & challenges.

For Roima Intelligence India Pvt Ltd.



Disha Thakkar
HR Department



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With sincere regards,
Kishan & Dev

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Chapter 1

Introduction

1.0 PROJECT DETAILS

- **Project Name:** Human Resource Management System (HRMS) – Internship Project
- **Application Type:** Web-Based Human Resource Management System
- **Domain:** Enterprise Workforce & Operations Management
- **Modules Covered:**
 - Travel & Expense Management
 - Social Achievements & Celebrations
 - Games Scheduling System
 - Organization Chart
 - Job Listing, Sharing & Referral System
 - Master Data Configuration (Employee, Role, Department)

1.1 PURPOSE

The HRMS application is designed as a centralized platform to streamline and automate internal organizational processes related to employee management, travel planning expense tracking, engagement activities, and recruitment workflows.

The system enables:

- Efficient **travel planning and expense reimbursement**
- Transparent **approval workflows**
- Enhanced **employee engagement through social features**
- Fair and automated **game scheduling system**
- Structured **organizational hierarchy visualization**
- Simplified **job sharing and referral process**

By integrating all these functionalities into a single platform, the system reduces manual effort, improves data accuracy, and enhances overall organizational productivity. The use of secure authentication, role-based access control, and cloud infrastructure ensures scalability, reliability, and data security.

1.2 SCOPE

The HRMS system aims to provide a scalable and modular solution for managing core HR operations in a modern organization. The scope includes the following major modules:

- Master Data Management
 - Employee Profile Management (CRUD operations)
 - Role & Permission Management

- Department Configuration
- Travel & Expense Module
 - Travel plan creation and assignment by HR
 - Document management (HR & Employee uploads)
 - Expense submission with validation rules and deadlines
 - Approval and verification workflow
- Social Engagement Module
 - Achievement posts, likes, and comments
 - Content moderation by HR
 - Automated birthday and work anniversary posts
- Games Scheduling Module
 - Slot booking for games (Foosball, Chess, Pool)
 - Fair allocation using queue-based scheduling
 - Weekly reset of participation cycle
- Organization Chart
 - Hierarchical visualization of employees
 - Managerial chain and direct reports
- Recruitment Module
 - Job listing and sharing via email
 - Employee referral system with CV upload
 - Referral tracking and audit logs

This system is designed to be extensible, allowing future enhancements such as analytics dashboards, reporting tools, and integrations with third-party services.

1.3 OBJECTIVE

The primary objectives of the HRMS system are:

- **Process Automation:** Reduce manual intervention in travel management, expense tracking, and recruitment workflows.
- **Improved Efficiency:** Enable faster approvals, better data handling, and streamlined operations.
- **Enhanced User Experience:** Provide an intuitive and responsive interface for employees, managers, and HR.
- **Data Accuracy & Validation:** Enforce business rules such as expense limits, deadlines, and mandatory fields.
- **Fair Resource Utilization:** Implement a balanced scheduling system for games using queue-based logic.

- **Security & Compliance:** Ensure secure access using authentication mechanisms and maintain audit logs for tracking system usage.
- **Scalability:** Build a cloud-based architecture that supports organizational growth and future feature expansion.

1.4 TECHNOLOGY REVIEW

[Frontend Technologies]

- **React (Vite + TypeScript):**

A modern frontend framework used to build a fast, scalable, and component-based user interface.

- **Tailwind CSS:**

Utility-first CSS framework for designing responsive and clean UI.

- **TanStack Query (React Query):**

Efficient data fetching, caching, and state synchronization between client and server.

- **Axios:**

Used for handling HTTP requests and API communication.

[Backend Technologies]

- **Java Spring Boot:**

Provides a robust backend framework for building RESTful APIs and handling business logic.

- **Spring Security with JWT Authentication:**

Implements secure authentication and role-based authorization using token-based security.

[Database & Cloud]

- **Azure MySQL Database:**

Cloud-based relational database for storing structured HRMS data.

- **Azure Blob Storage:**

Used for storing documents such as travel files, receipts, and CV uploads securely.

[Development Tools]

- **IntelliJ IDEA:** Backend development environment

- **Visual Studio Code:** Frontend development

- **SSMS / Database Tools:** Data visualization and management

Chapter 2

About the system

2.0 STUDY OF CURRENT SYSTEM

The existing system in the organization is a basic Employee Management System that primarily handles core HR functionalities such as leave management, attendance tracking, and holiday management. While this system provides essential support for daily HR operations, it lacks several advanced features required for modern organizational needs.

The current system does not support modules such as travel planning, expense reimbursement, employee engagement, game scheduling, organizational visualization, or recruitment workflows. Many of these processes are either handled manually or through disconnected tools like emails and spreadsheets, leading to inefficiencies and lack of transparency.

To overcome these limitations, the proposed HRMS system is designed as a comprehensive, cloud-based solution that extends the capabilities of the existing system by introducing multiple integrated modules, improving automation, and enhancing user experience.

2.1 PROBLEM AND WEAKNESS OF CURRENT SYSTEM

The current system faces the following major limitations:

- **Limited Functionality:** Only supports leave, attendance, and holiday management, lacking advanced HR features like travel, expenses, and recruitment.
- **Manual Processes:** Travel planning, expense submission, and approvals are handled manually, increasing chances of errors and delays.
- **No Document Management:** No structured system to upload, manage, and retrieve important documents like receipts, tickets, or CVs.
- **Lack of Transparency:** Employees cannot track the status of requests (expenses, approvals, referrals) in real time.
- **No Employee Engagement Features:** Missing social interaction modules such as achievements, celebrations, and internal communication.
- **Inefficient Resource Utilization:** No system for fair scheduling of shared resources like games.
- **No Recruitment Workflow:** Job sharing and employee referral processes are not systemized.
- **Weak Audit & Security Tracking:** No proper logging of user activities such as login history, IP tracking, or action audits.

2.2 USER CHARACTERISTICS

The system is designed for three primary user roles:

- Employees
 - Basic users of the system
 - Submit expenses, upload documents, book game slots, and interact with social features
 - Require a simple, responsive, and user-friendly interface
- HR / Admin
 - Full system control and configuration
 - Manage employees, travel plans, job postings, and approvals
 - Require advanced dashboards, filters, and administrative controls
 - Need secure access with role-based permissions

2.3 HARDWARE AND SOFTWARE REQUIREMENTS

2.3.1 Hardware Requirements

- Minimum Intel i5 / equivalent processor
- 8 GB RAM (recommended for smooth performance)
- Stable internet connection
- Cloud server for deployment (Railway / Azure-based infrastructure)

2.3.2 Hardware Requirements

- **Frontend:**
 - Web Browser (Chrome, Edge, Firefox)
 - React (Vite + TypeScript)
 - Tailwind CSS for UI styling
- **Backend:**
 - Java 17
 - Spring Boot Framework
 - Spring Security with JWT Authentication
 - Spring Data JPA & JDBC
 - Spring Mail for email notifications
 - WebSocket for real-time updates
- **Database & Storage:**
 - Azure MySQL Database
 - Azure Blob Storage for file handling
- **Development & Tools:**
 - IntelliJ IDEA (Backend Development)
 - Visual Studio Code (Frontend Development)
 - Postman (API Testing)

- SSMS / DB Tools (Data visualization)
- GitHub (Version Control)
- Railway (Deployment Platform)

2.4 CONSTRAINTS

2.4.1 Regulatory Policies

- Follow standard coding practices for Java Spring Boot and React
- Maintain proper API documentation using OpenAPI/Swagger
- Use Git for version control with structured commits
- Ensure proper validation and error handling

2.4.2 Hardware Limitations

- Performance depends on internet speed and server availability
- Large file uploads (documents, CVs) may impact response time if not optimized

2.4.3 Interfaces with Other Applications

- Email Services (SMTP) for notifications and alerts
- Cloud Storage (Azure Blob) for document management
- Future scope: Integration with calendar APIs and HR tools

2.4.4 Parallel Operations

The system supports multiple concurrent users:

- Employees submitting expenses and interacting with feeds
- HR managing approvals and configurations
- Real-time updates using WebSockets
- REST APIs handling concurrent requests efficiently

2.4.5 Reliability Requirements

To ensure system stability:

- High availability through cloud deployment (Railway)
- Proper exception handling and logging
- Database consistency

2.4.6 Criticality of the Application

The HRMS system is a critical internal tool that directly impacts:

- Financial processes (expense management)
- Employee satisfaction (engagement features)
- Operational efficiency (automation of workflows)

2.4.7 Safety and Security Considerations

- JWT-based authentication and authorization
- Role-based access control (RBAC)
- Secure API endpoints with Spring Security
- Encrypted data transmission (HTTPS)
- Login audit logging (IP, timestamp, user-agent)
- Secure file storage using cloud services

2.5 ASSUMPTIONS AND DEPENDENCIES

- The system assumes continuous internet connectivity for users
- Cloud services (Railway, Azure) are available and stable
- Email services are properly configured for notifications
- Users follow defined workflows (e.g., expense deadlines, booking rules)
- Database and backend services maintain high uptime

Chapter 3

System Analysis

3.0 Overview

The System Analysis chapter examines the HRMS application's functionalities, user interactions, and system behavior. It outlines key use cases and component interactions, clarifying requirements and design. The HRMS streamlines employee processes such as travel and expense management, social engagement, game scheduling, organizational charts, and referrals, providing a clear view of how the system supports organizational objectives.

3.1 Use Case Diagram



[Fig 3.1.1 Use Case Diagram - HRMS]

The use case diagram illustrates the interactions between users (Employee and HR/Admin) and the HRMS system.

Actors

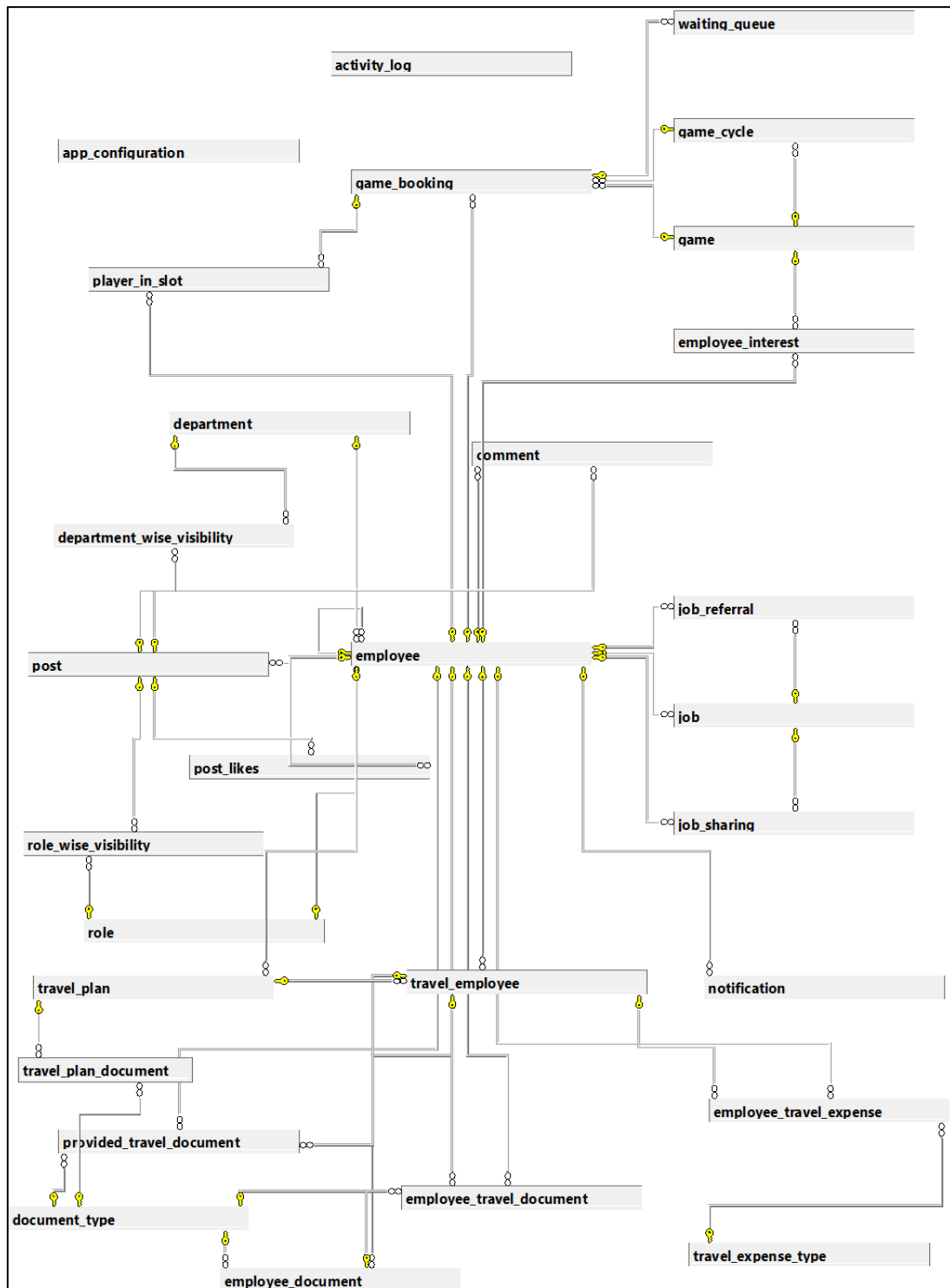
- **Employee:** A user who views assigned travels, submits expenses, participates in social achievements, books games, views organizational charts, and shares or refers jobs.
- **HR/Admin:** A user who manages employee data, configures system settings, creates and approves travels and expenses, moderates social content, schedules games, and oversees job postings and referrals.

Use Cases

1. **Login / Logout:** Authenticate users based on role (Employee or HR/Admin).
2. **Configure System Settings:** HR/Admin can manage core system configurations.
3. **Manage Employee Data:** HR/Admin can create, read, update, and delete employee profiles.
4. **Manage Role Data:** HR/Admin can define and manage roles and permissions.
5. **Manage Department Data:** HR/Admin can create and manage organizational departments.
6. **Create Travel Plan:** HR/Admin assigns travel to employees and triggers notifications.
7. **View Assigned Travels:** Employees can view their travel plans and related details.
8. **Upload/View Travel Documents:** Employees can upload/view their own travel documents; HR/Admin can upload/view all.
9. **Submit Expense:** Employees submit expenses linked to travel plans, including proof documents.
10. **Approve/Reject Expense:** HR/Admin verifies and updates expense status with optional remarks.
11. **View Expenses:** Employees can track expense status and HR remarks.
12. **Create Achievement Post:** Employees share accomplishments in the achievements feed.
13. **View Achievement Feed:** Employees and HR/Admin can see posts, like, and comment.
14. **Edit/Delete Own Post/Comment:** Employees can manage their own content.
15. **Moderate Content:** HR/Admin can delete inappropriate posts/comments and notify the employee.
16. **Configure Game Slots:** HR/Admin sets up games, operating hours, and slot parameters.
17. **Book Game Slot:** Employees book game slots following fairness rules.
18. **Cancel Game Booking:** Employees can cancel, and next eligible player is reassigned.
19. **View Organization Chart:** Employees and HR/Admin can browse hierarchical relationships.
20. **Share Job via Email:** Employees can share jobs with external contacts.
21. **Refer Friend:** Employees refer friends by submitting their details and CV; HR/Admin is notified.

3.2 Class Diagram

The class diagram represents the static structure of the HRMS (Human Resource Management System) application, illustrating the key entities, their relationships, and overall system organization. It acts as a blueprint for understanding how different modules such as travel management, expense tracking, social engagement, game scheduling, and job referrals interact within the system.



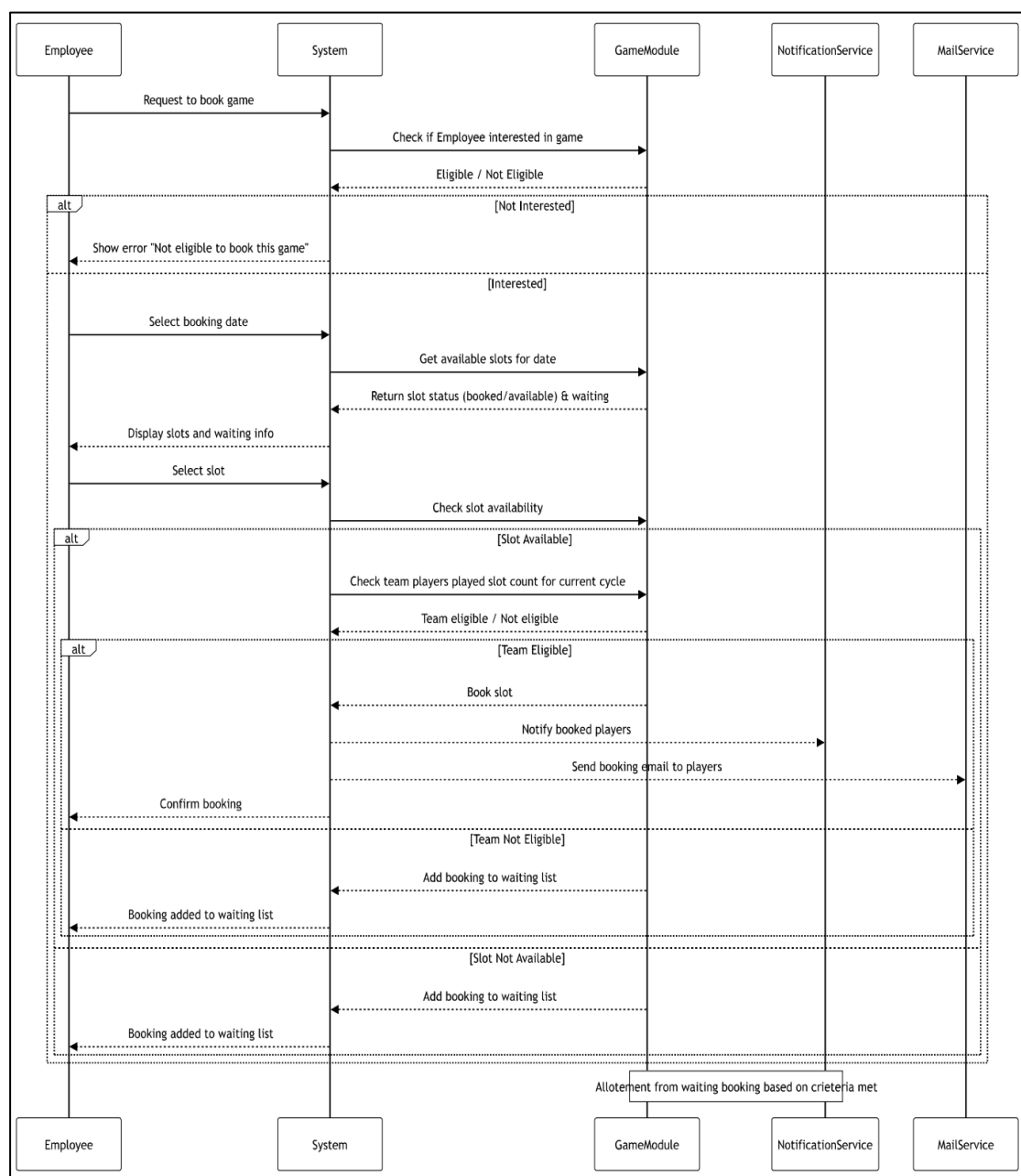
[Fig 3.2.1 Class Diagram - HRMS]

3.3 Sequence Diagram

3.3.1 Game Booking Sequence Diagram

The sequence diagram shows the interaction between the employee and system during the game booking process. It begins with the employee requesting a booking, after which the system verifies the employee's interest in the selected game. The system then displays available slots along with booking and waiting status.

Based on slot availability and fairness rules (players' participation in the current cycle), the booking is either confirmed or added to the waiting list. The system also manages waiting bookings through cancellations and eligibility checks, while sending notifications and emails to all involved users.

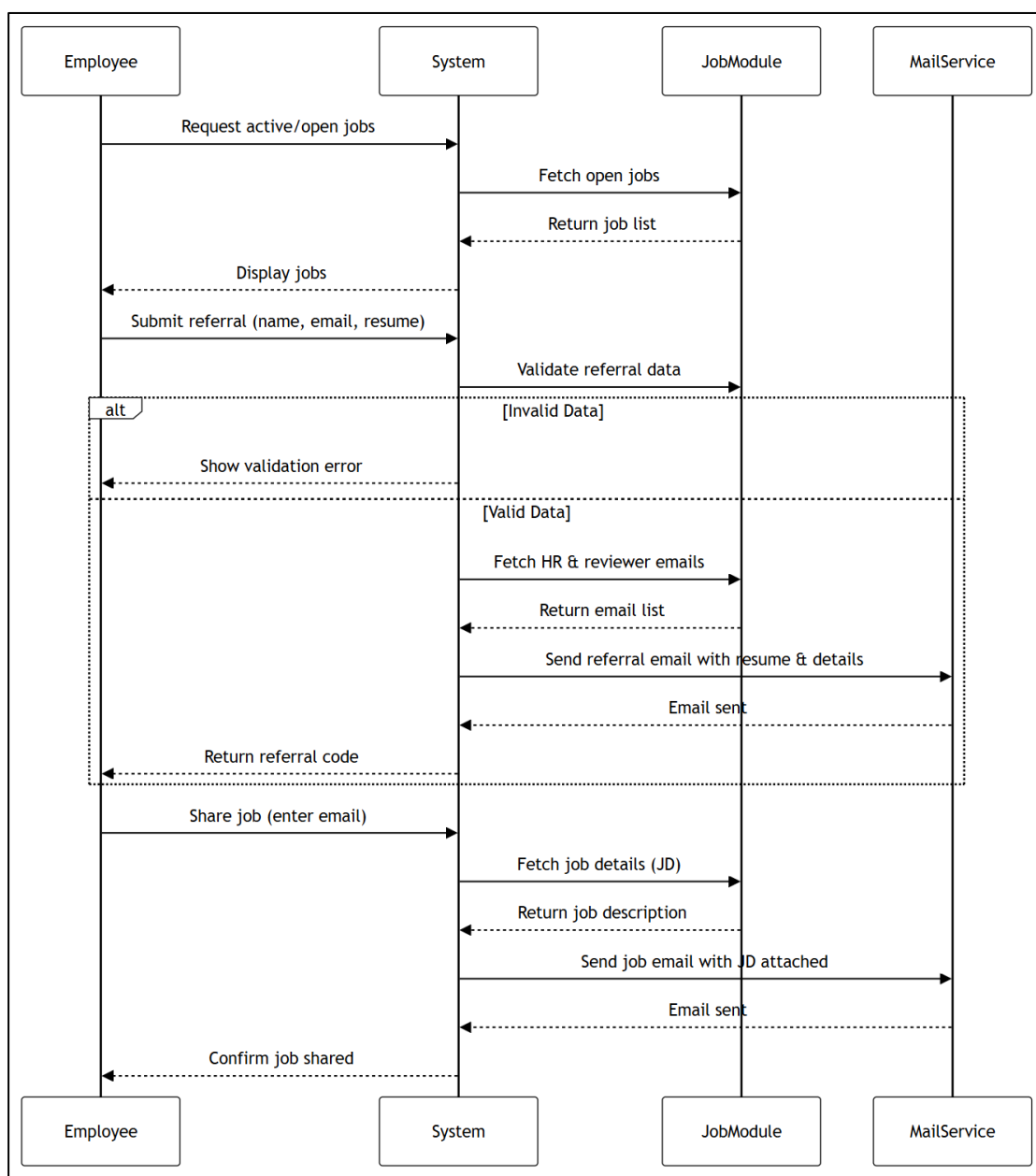


[Fig 3.3.1 Sequence Diagram – Game Booking Module]

3.3.2 Job Referral & Job Sharing Sequence Diagram

The sequence diagram illustrates the workflow for job referral and job sharing. The employee first views active job openings and submits referral details, including candidate information and resume. The system validates the data, retrieves relevant HR emails, and sends the referral details via email. A referral code is then generated for tracking.

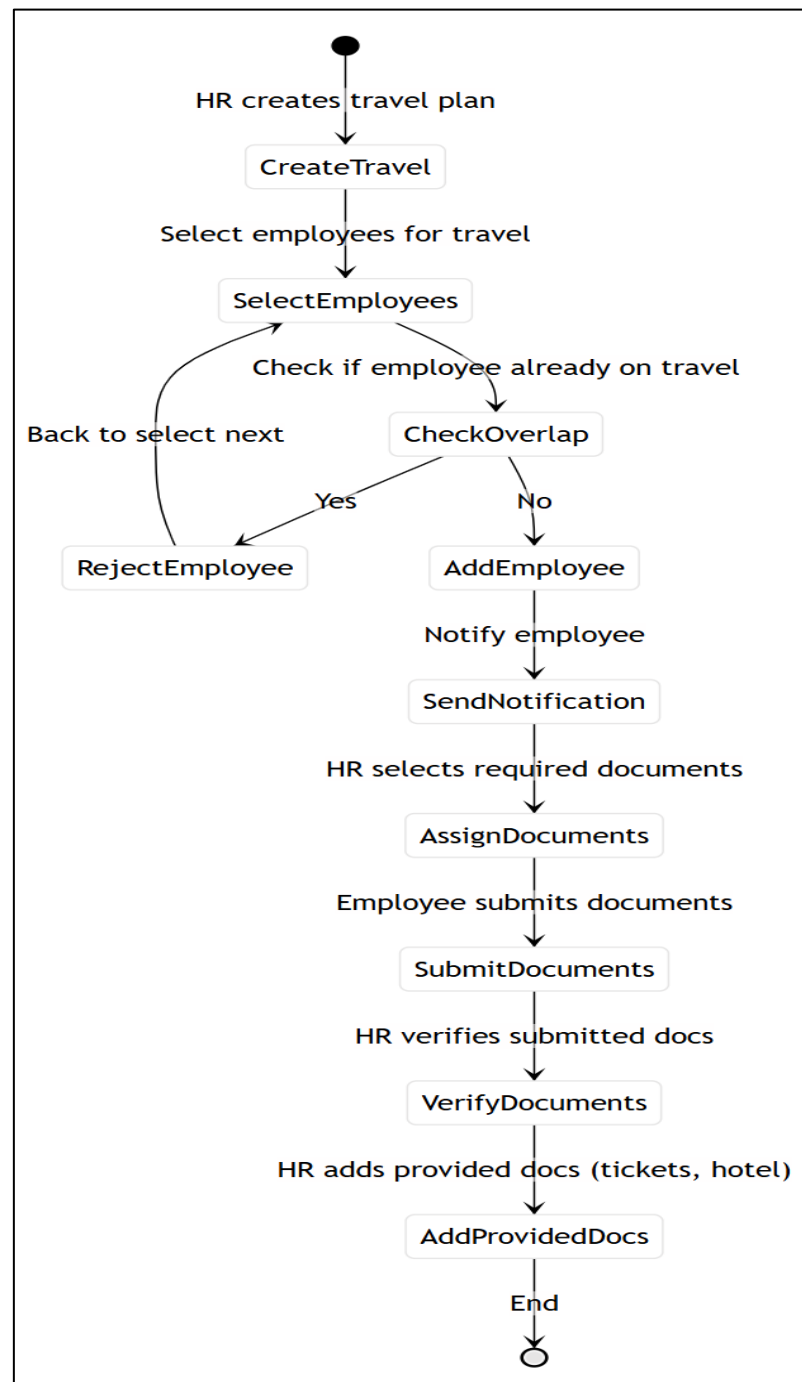
For job sharing, the employee provides recipient email addresses, and the system sends job details along with the job description. This ensures efficient communication of job opportunities.



[Fig 3.3.2 Sequence Diagram – Job Sharing & Referral Module]

3.4 Activity Diagram

The activity diagram illustrates the workflow of the Travel Plan module in the HRMS system. It starts with HR creating a travel plan with a specified date range. The HR selects employees for the travel; the system checks for overlapping travel plans, and only employees without conflicts are added. Notifications are sent to selected employees. Next, HR assigns the required documents for the travel, which employees submit for verification. HR reviews and approves the submitted documents. Finally, HR adds provided documents such as flight tickets or hotel confirmations to complete the travel plan workflow.



[Fig 3.4.1 Activity Diagram – Travel Plan Module]

3.5 Functional Requirements

Employees

R.1 Travel Management

- **Description:** Employees can view assigned travels and related documents.
 - **R.1.1 Viewing Travel Details**
 - **Input:** Employee selects a travel plan.
 - **Output:** Travel details, including dates, locations, and purpose, are displayed.
 - **R.1.2 Uploading Travel Documents**
 - **Input:** Employee uploads documents (visa, boarding pass, tickets).
 - **Output:** Documents are linked to the selected travel and visible to HR.

R.2 Expense Management

- **Description:** Employees submit and track expenses associated with travel.
 - **R.2.1 Submitting Expense**
 - **Input:** Employee enters expense details, selects travel, and uploads proof.
 - **Output:** Expense is saved in draft or submitted for approval.
 - **R.2.2 Viewing Expense Status**
 - **Input:** Employee accesses their expense list.
 - **Output:** Status, HR remarks, and total claimed amount are displayed.

R.3 Social Engagement

- **Description:** Employees interact with achievements, birthdays, and anniversaries.
 - **R.3.1 Creating Achievement Post**
 - **Input:** Employee enters post title, description, and tags.
 - **Output:** Post appears in the internal feed with like and comment options.
 - **R.3.2 Interacting with Posts**
 - **Input:** Employee likes or comments on a post.
 - **Output:** Engagement is recorded and visible to other users.

R.4 Game Scheduling

- **Description:** Employees book time slots for organizational games.
 - **R.4.1 Requesting Game Slot**
 - **Input:** Employee selects game type and preferred slot.
 - **Output:** Slot is assigned based on fairness rules, and notifications are sent.

- **R.4.2 Canceling Slot**
 - **Input:** Employee cancels a booked slot.
 - **Output:** Slot is freed and reassigned to the next eligible player.

R.5 Job Referral

- **Description:** Employees can share job openings and refer friends.
 - **R.5.1 Sharing Job**
 - **Input:** Employee enters recipient email(s).
 - **Output:** Job details are sent via email.
 - **R.5.2 Referring Friend**
 - **Input:** Employee enters friend details and uploads CV.
 - **Output:** Referral is logged and email notification is sent to HR.

HR

R.1 Employee & System Configuration

- **Description:** HR manages employees, roles, and master data.
 - **R.1.1 Managing Employee Profiles**
 - **Input:** HR adds, updates, or deletes employee information.
 - **Output:** Employee records are updated and accessible based on role.
 - **R.1.2 Configuring System Roles**
 - **Input:** HR defines roles and permissions.
 - **Output:** Role-based access is enforced across modules.
 - **R.1.3 Department Management**
 - **Input:** HR adds or updates department details.
 - **Output:** Departments are reflected in org chart and workflows.

R.2 Travel & Expense Management

- **Description:** HR assigns travel plans, manages documents, and approves expenses.
 - **R.2.1 Creating Travel Plan**
 - **Input:** HR selects employees, travel dates, and details.
 - **Output:** Employees receive email and in-app notifications.
 - **R.2.2 Managing Travel Documents**
 - **Input:** HR uploads travel-related files.
 - **Output:** Documents are linked to travel and visible to employees and managers.
 - **R.2.3 Approving Expenses**
 - **Input:** HR reviews expense details and proof.
 - **Output:** Expense status is updated to Approved or Rejected, with remarks.

R.3 Social Engagement & Recognition

- **Description:** HR monitors posts and manages content moderation.
 - **R.3.1 Deleting Inappropriate Content**

- **Input:** HR selects posts or comments to remove.
- **Output:** Content is deleted and email notification sent to the concerned employee.

R.4 Games Scheduling Configuration

- **Description:** HR configures games, slots, and weekly schedules.
 - **R.4.1 Configuring Slots**
 - **Input:** HR sets game types, duration, and max players.
 - **Output:** Scheduler runs weekly and ensures fair rotation of players.

R.5 Job & Referral Management

- **Description:** HR manages job postings and referrals.
 - **R.5.1 Configuring Job Openings**
 - **Input:** HR creates job details and sets CV reviewers.
 - **Output:** Job is visible to employees and shareable.
 - **R.5.2 Reviewing Referrals**
 - **Input:** HR views referrals and tracks their status.
 - **Output:** Referral status is updated and audit logs maintained.

3.6 Other Non-Functional Requirements

1. Performance Requirements

- The system must support multiple concurrent users with minimal delay.
- Travel and expense data should load within 2–3 seconds.

2. Security Requirements

- Role-based authentication (Employee / HR).
- JWT tokens for secure API access.
- Data encryption for sensitive information.

3. Usability Requirements

- Clean, responsive UI built with React, Tailwind, and Vite.
- Clear navigation for employees and HR with minimal learning curve.

4. Reliability & Audit

- System maintains login and activity logs (timestamp, IP, device).
- Cloud deployment ensures high availability and backups.

Chapter 4

Design

4.1 Database Design

EMPLOYEE

Description: Stores employee information including personal, departmental, and role details.

[Table 4.1.1 Employee Schema]

Name	Data Type	Allow Null
pk_employee_id	int	False
date_of_birth	date	True
email	varchar(255)	False
first_name	varchar(255)	True
joining_date	date	True
last_name	varchar(255)	True
password	varchar(255)	False
fk_department_id	int	True
fk_manager_employee_id	int	True
fk_role_id	int	True
is_deleted	bit	True

DEPARTMENT

Description: Maintains organizational departments to which employees are assigned.

[Table 4.1.2 Department Schema]

Name	Data Type	Allow Null
pk_department_id	int	False
department_name	varchar(255)	False

ROLE

Description: Defines employee roles (like HR, Admin, Staff) for role-based access control in the system.

[Table 4.1.3 Role Schema]

Name	Data Type	Allow Null
pk_role_id	int	False
role_name	varchar(255)	False

TRAVEL_PLAN

Description: Contains travel plans created by employees, including start/end time and description.

[Table 4.1.4 Travel Plan Schema]

Name	Data Type	Allow Null
description	varchar(255)	False
end_time	datetime2(6)	True
is_active	bit	True
start_time	datetime2(6)	False
title	varchar(255)	False
fk_created_by_employee_id	int	False

EMPLOYEE_TRAVEL_EXPENSE

Description: Stores travel expenses submitted by employees, linked to specific travel plans, with approval workflow.

[Table 4.1.5 Employee Travel Expense Schema]

Name	Data Type	Allow Null
pk_employee_travel_expense_id	int	False
amount	float	False
created_at	date	True
expense_date	date	False
expense_detail	varchar(255)	True
proof_url	varchar(255)	True
remark	varchar(255)	True
status	smallint	True
updated_at	date	True
fk_approver_employee_id	int	True
fk_travel_employee_id	int	False
fk_travel_expense_type_id	int	False

GAME

Description: Contains game information for employee engagement activities.

[Table 4.1.6 Game Schema]

Name	Data Type	Allow Null
pk_game_id	int	False
duration_in_minute	int	False
end_time	time(7)	True
game_name	varchar(255)	True
is_deleted	bit	True
max_player	int	False
start_time	time(7)	True

JOB

Description: Stores job postings created by employees, including job details and status.

[Table 4.1.7 Job Schema]

Name	Data Type	Allow Null
pk_job_id	int	False
is_open	bit	True
job_description_url	varchar(255)	True
location	varchar(255)	True
opened_at	date	True
requirement	int	False
salary	int	False
title	varchar(255)	True
fk_created_by_employee_id	int	True

POST

Description: Represents posts shared by employees, either public or within departments/roles, supporting engagement features like comments and likes.

[Table 4.1.8 Post Schema]

Name	Data Type	Allow Null
pk_post_id	int	False
created_at	datetime2(6)	True
description	varchar(255)	True
is_active	bit	True
is_public	bit	True
is_system_generated	bit	True
title	varchar(255)	True
fk_author_employee_id	int	True
tags	varchar(255)	True

NOTIFICATION

Description: Manages notifications sent to employees regarding approvals, updates, or system events.

[Table 4.1.9 Notification Schema]

Name	Data Type	Allow Null
notification_id	int	False
created_at	datetime2(6)	True
is_read	bit	True
message	varchar(255)	True
notification_type	smallint	True
fk_employee_id	int	True

EMPLOYEE_DOCUMENT

Description: Stores official documents uploaded by employees, such as ID proofs, certificates, or any HR-required documents. Each document is linked to a document type for classification.

[Table 4.1.10 Employee Document Schema]

Name	Data Type	Allow Null
pk_employee_document_id	int	False
document_url	varchar(255)	False
uploaded_at	date	False
fk_document_type_id	int	False
fk_employee_id	int	False

EMPLOYEE_TRAVEL_DOCUMENT

Description: Manages documents submitted by employees during travel plans, including approvals and status tracking by the HR or manager.

[Table 4.1.11 Employee Travel Document Schema]

Name	Data Type	Allow Null
pk_employee_travel_document_id	int	False
action_date	date	True
document_status	smallint	True
remark	varchar(255)	True
fk_approver_employee_id	int	True
fk_document_type_id	int	False
fk_employee_document_id	int	True
fk_travel_employee_id	int	False

GAME_BOOKING

Description: Records employee bookings for games, including booking status, selected game, and timing, supporting scheduling and participation tracking.

[Table 4.1.12 Game Booking Schema]

Name	Data Type	Allow Null
pk_game_booking_id	int	False
booking_date	date	True
booking_status	smallint	True
booking_time	time(7)	True
created_at	datetime2(6)	True
fk_booked_by_employee_id	int	True
fk_game_id	int	True
fk_game_cycle_id	int	True

JOB_REFERRAL

Description: Stores employee referrals for job openings, including referee details, resume links, referral status, and the employee who referred. This helps track internal recruitment recommendations.

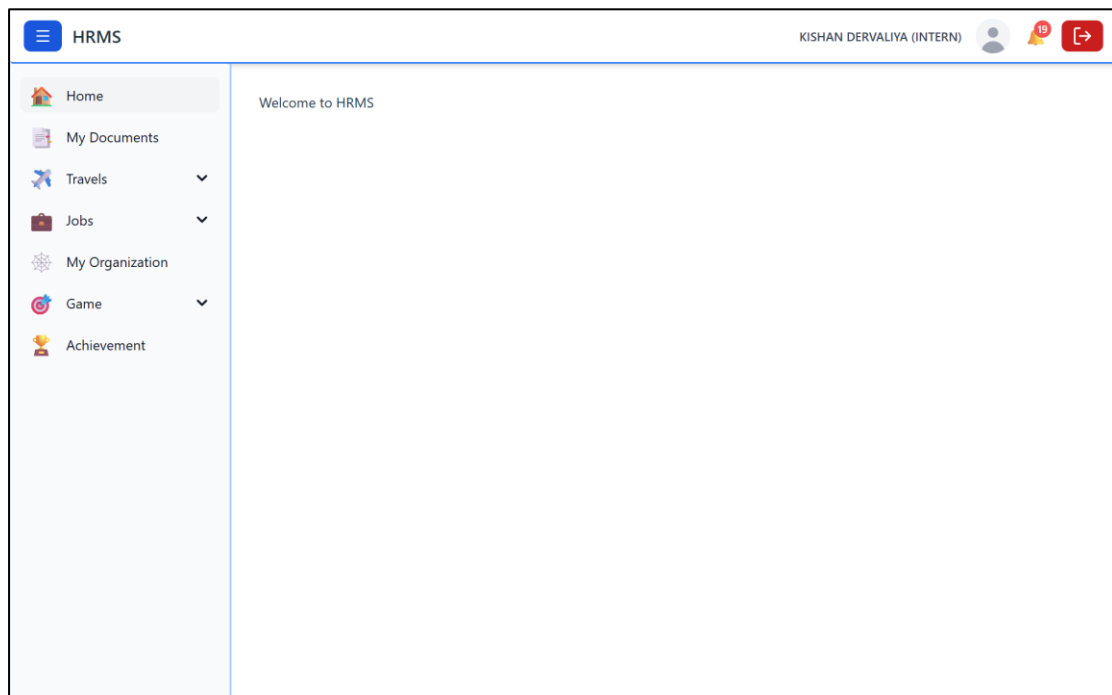
[Table 4.1.13 Job Referral Schema]

Name	Data Type	Allow Null
pk_job_referral_id	uniqueidentifier	False
referee	varchar(255)	True
referee_email	varchar(255)	True
referral_status	smallint	True
resume_url	varchar(255)	True
fk_job_id	int	True
fk_referrer_employee_id	int	True

4.2 Frontend Interface

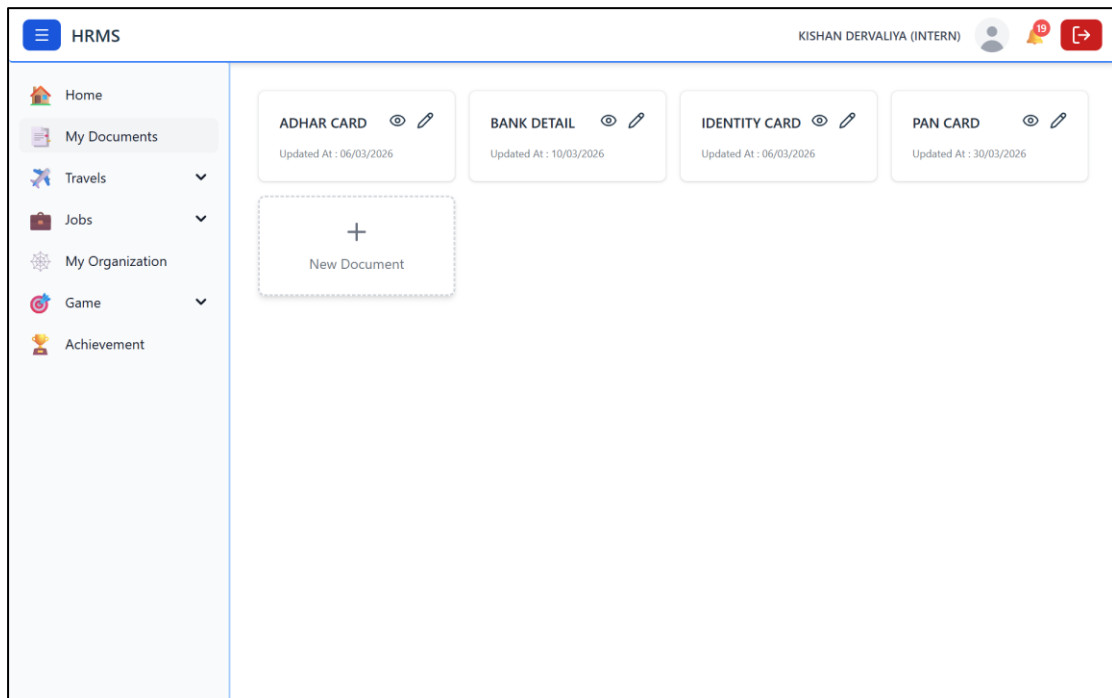
The front-end of the HRMS system is a web-based interface built with ReactJS and styled with TailwindCSS, providing a responsive and intuitive design for employees and HR users. It supports real-time updates via WebSocket for notifications and activity tracking.

- Home page Showing sidebar and navbar view, on navbar username, notification count and logout button listed. On sidebar different navigation is listed.



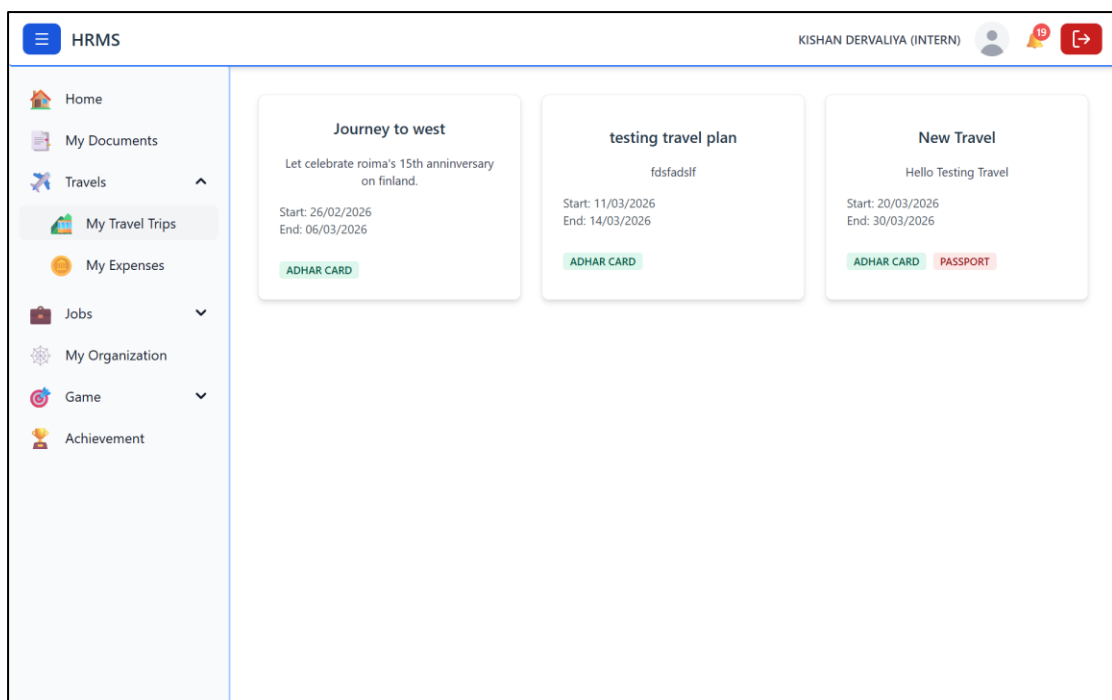
[Fig 4.2.1 Home Page]

- Document Page shows all document of employee



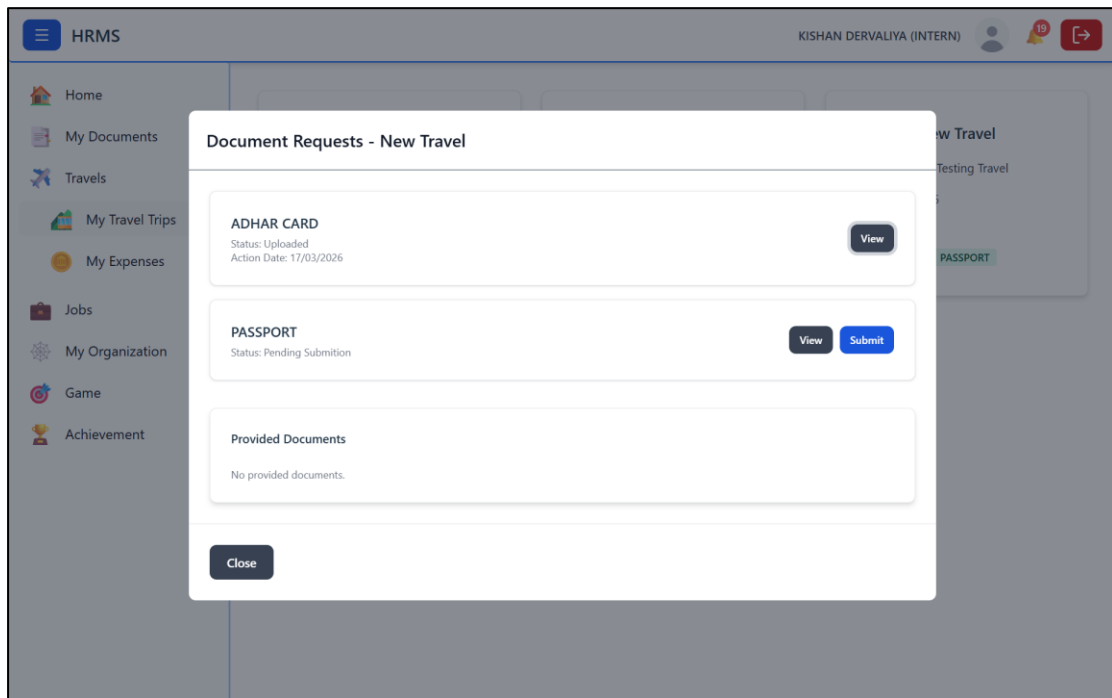
[Fig 4.2.2 Document Page]

- Travel Trips Shows all Travel Plan in which employee selected. Here documents also listed which need to bring with plan.



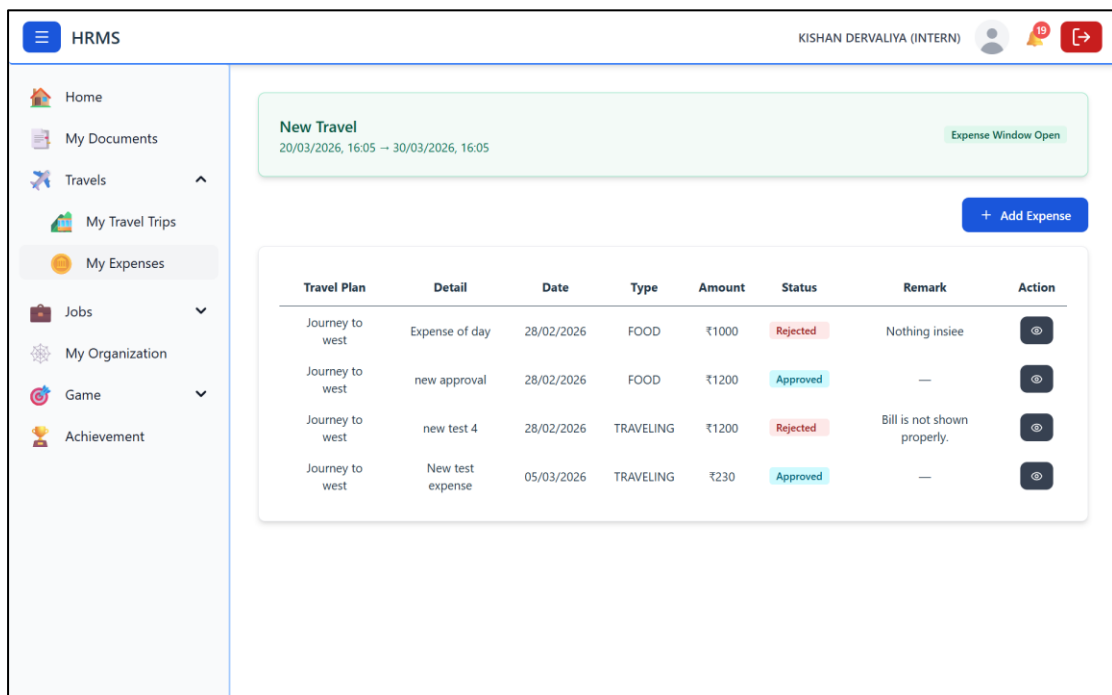
[Fig 4.2.3 My Travel Plan Page]

- Document Submission Window for employee



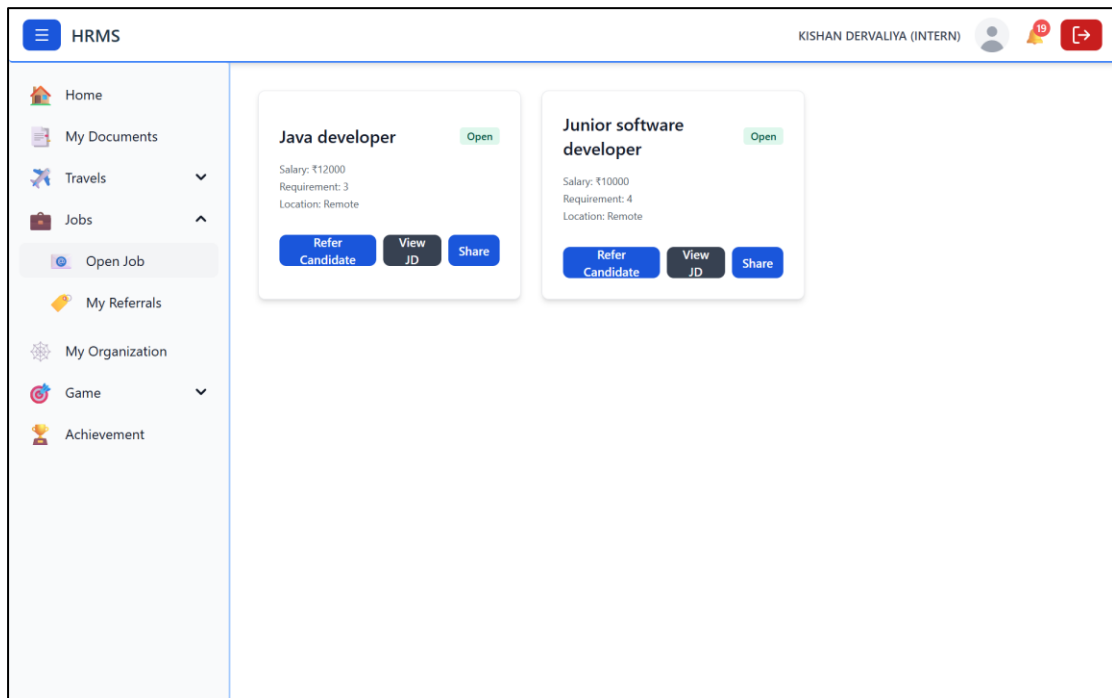
[Fig 4.2.4 Document Submission Page]

- Travel Expense Page shows all approved, rejected, draft expense request.



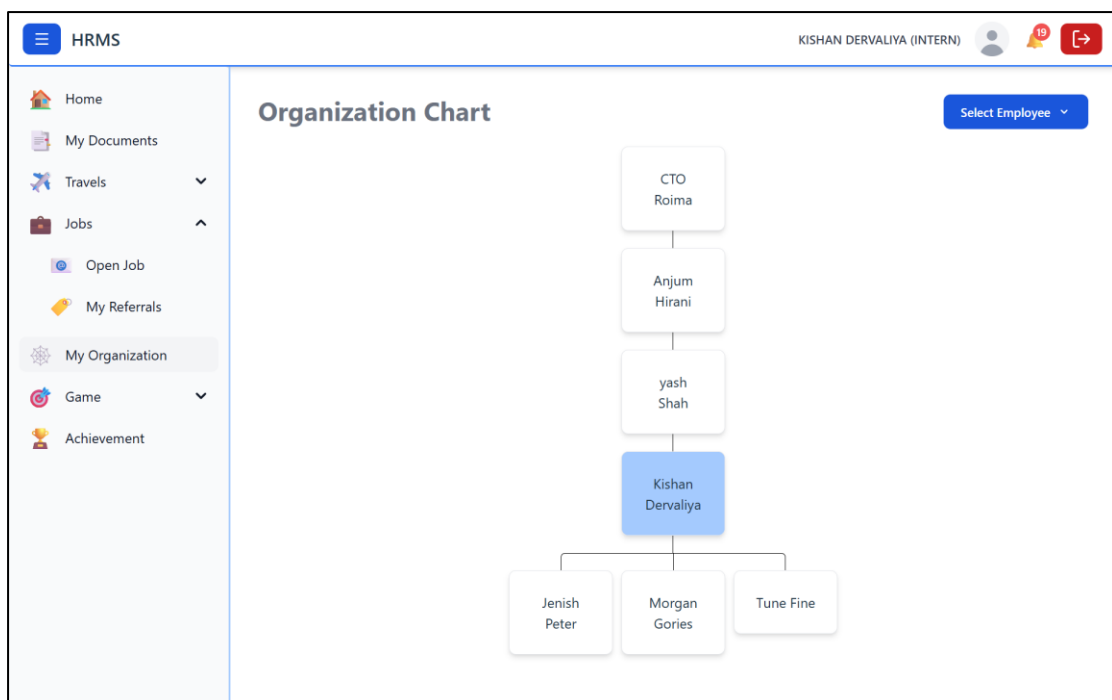
[Fig 4.2.5 Expense Submission Page]

- Open Job page show all open position details with refer option & share option



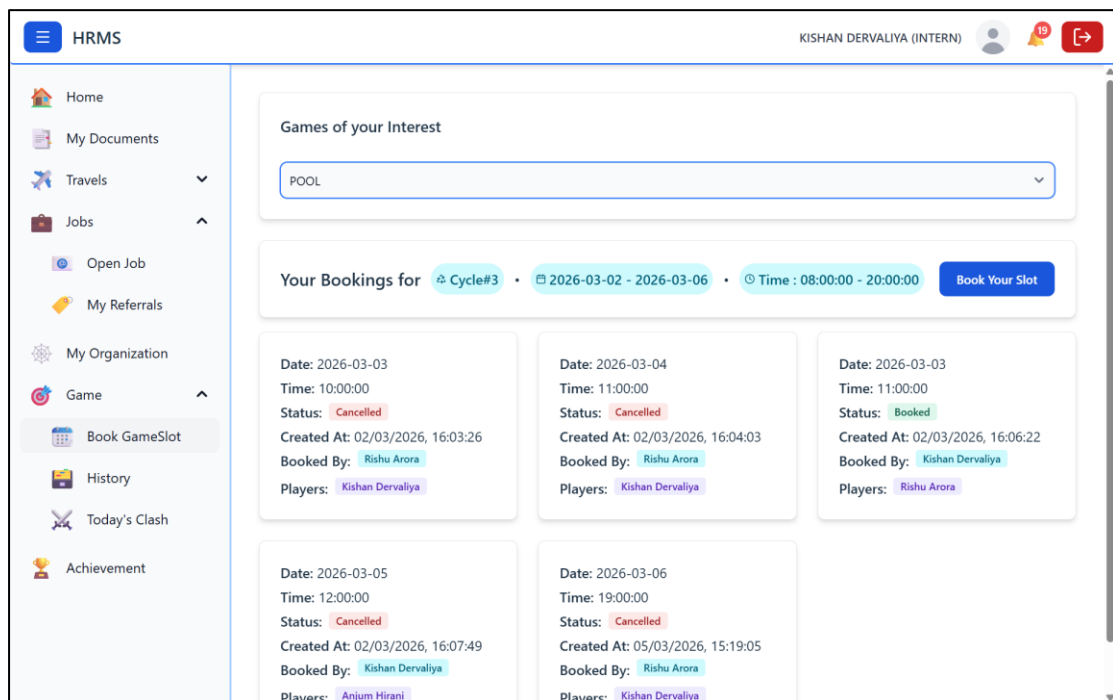
[Fig 4.2.6 Job Position Page]

- Organization chart page for all employee, show tree like structure for employee supervisor hierarchy



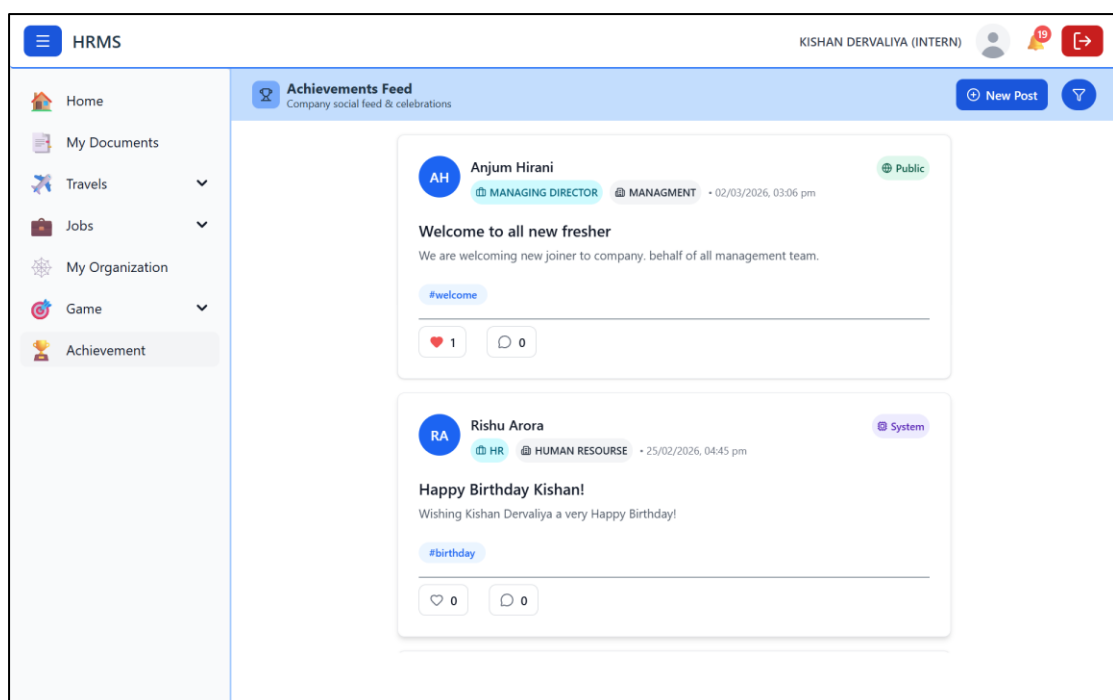
[Fig 4.2.7 Organization Chart Page]

- Game booking page for showing all booking and current cycle information



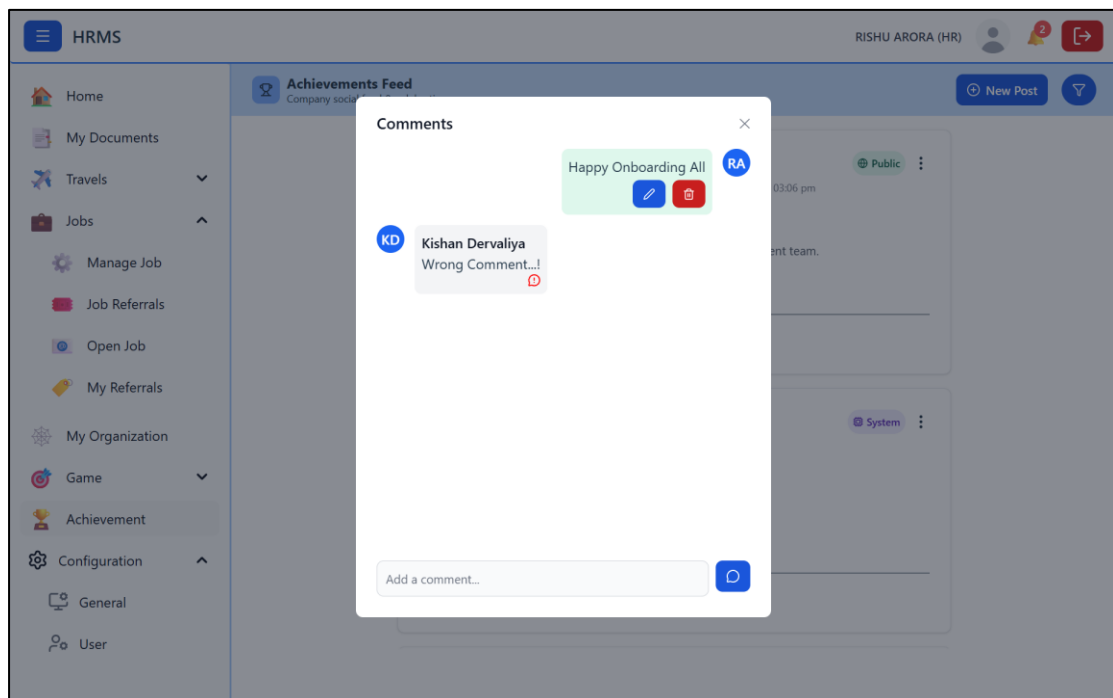
[Fig 4.2.8 Game Booking Page]

- Achievement Feed page for sharing and view post related company achievement.



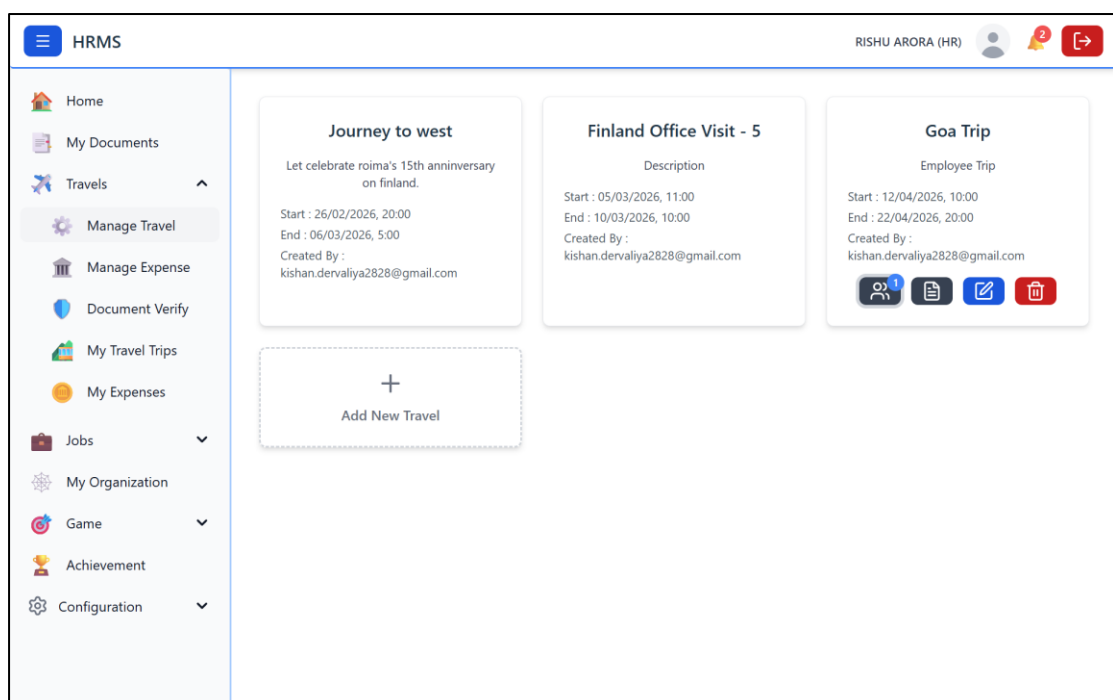
[Fig 4.2.9 Achievement Feed Page]

- Chat section on feed with edit and delete option, hr also can delete moderated comment from chat.



[Fig 4.2.10 Chat Page]

- Manage Travel page for hr where they can manage travel plan and employee selectionn



[Fig 4.2.11 Manage Travel Page]

- General Configuration page for app params of system (Hr authorized page)

[Fig 4.2.12 General configuration Page]

- Employee Data modification & management page

Role ID	Role Name	Actions
2	DEVELOPER	
3	HR	
4	INTERN	
5	JUNIOR DEVELOPER	
6	MANAGING DIRECTOR	
7	TEAMLEAD	

New Role

Employees

Filter by Role Filter by Department

Email	DOB	Joining	Department	Role	Manager	Actions
an.dervaliya2828@gmail.com	19/03/2002	19/02/2026	HUMAN RESOURCE	HR	Anjum Hirani	
han.dervaliya@roimaint.com	25/02/2002	19/02/2026	TRAINING	INTERN	yash Shah	

[Fig 4.2.13 User Configuration Page]

4.3 Navigation

- **Top Navigation Bar:** Quick links to Dashboard, Notifications, Profile, and Logout.
- **Sidebar Menu:** Role-based menu items; for employees: Travel, Games, Referrals, Posts; for HR: Employee Management, Travel Approvals, Job Posting.
- **Breadcrumbs:** Shows current module location to help users navigate back easily.
- **Responsive Design:** Adjusts layout for desktops, tablets, and mobile screens.

4.4 Validations

1. **Client-Side Validation** (Front-End)
 - Done using **React Hook Form** and custom logic.
 - Ensures required fields are filled, correct formats (email, date, number), and file types for document uploads.
 - Example: Employee cannot submit travel request without start/end date or travel document.
2. **Server-Side Validation** (Back-End)
 - Spring Boot APIs validate incoming data for security and business rules.
 - Checks for **duplicate entries**, **foreign key references**, and **role-based permissions**.
 - Example: Employee travel expense cannot exceed the maximum allowed limit defined in TravelExpenseType.

Chapter 5

Implementation

5.0 Implementation Planning

The Human Resource Management System (HRMS) is implemented as a modular, scalable web-based solution integrating key HR functions such as Travel & Expense, Social Engagement, Game Scheduling, Organization Chart, and Recruitment. Development begins with requirement analysis based on user roles (Employee, Manager, HR/Admin) and follows a layered architecture consisting of frontend, backend, and database components to ensure maintainability and parallel development. An Agile methodology is adopted for iterative development and continuous improvement, covering phases like database design, backend API development using Spring Boot, frontend development using React (Vite + TypeScript), integration via REST APIs, and testing. Key features include Role-Based Access Control (RBAC), JWT-based authentication, audit logging, email notifications, and document storage, with final deployment on a cloud platform to ensure scalability, reliability, and security.

5.1 Implementation Environment

Tools and Technologies Used

- Frontend Technologies
 - React (Vite + TypeScript)
 - Tailwind CSS
 - TanStack Query (React Query)
 - Axios
- Backend Technologies
 - Java Spring Boot
 - Spring Security with JWT Authentication
- Database & Cloud
 - Azure MySQL Database
 - Azure Blob Storage

System Characteristics

Multi-user System:

- The HRMS is designed as a multi-user system supporting Employees, Managers, and HR/Admin users simultaneously.
- Ensures real-time updates across modules such as expense approvals, social feeds, and scheduling.

GUI vs Non-GUI:

- The frontend is fully GUI-based with responsive and interactive components such as dashboards, forms, and feeds.
- The backend operates as a non-GUI system handling APIs, business logic, authentication, validation, and logging.

5.2 Program / Module Specification

The system is implemented using a modular architecture to ensure scalability and maintainability. Each module is independently developed and integrated via REST APIs.

Master Data Configuration Module: Manages core organizational data including employees, roles, permissions, departments, and system settings. It ensures proper configuration and control through RBAC and customizable validation rules.

Travel & Expense Management Module: Handles travel planning, expense submissions, approvals, and document management. It includes validation rules, deadline controls, status tracking, and notification support.

Social Achievements & Engagement Module: Enables employees to share achievements, interact through likes and comments, and receive automated posts for events like birthdays and anniversaries, with HR moderation for content.

Games Scheduling Module: Facilitates fair game slot booking based on player interest, capacity, and participation history. It includes queue-based allocation, weekly resets, and notification integration.

Organization Chart Module: Provides a hierarchical view of the company structure, allowing users to explore reporting relationships and navigate between employee roles interactively.

Job Listing, Sharing & Referral Module: Supports job postings, sharing opportunities, and employee referrals with CV uploads and tracking. It includes notifications and audit logs for recruitment activities.

Authentication & Audit Logging Module: Ensures secure system access using JWT and RBAC, while maintaining detailed logs of user activities, login details, and system actions for monitoring and compliance.

Architecture Overview

- **Frontend Layer:** React-based UI with reusable components and state management using React Query
- **Backend Layer:** Spring Boot REST APIs with controllers, services, and repositories
- **Database Layer:** Azure MySQL storing structured HRMS data
- **Storage Layer:** Azure Blob Storage for document handling
- **Security Layer:** JWT-based authentication and authorization

5.3 CODING STANDARDS

To ensure maintainability and code quality, the following coding standards are followed:

Naming Conventions

- camelCase for variables and functions
- Meaningful and descriptive naming for APIs and database entities

Code Formatting

- Consistent formatting using Prettier and ESLint (frontend)
- Standard Java conventions for backend

Error Handling

- Try-catch blocks in backend services
- Standardized API response structure
- User-friendly error messages

Validation

- Server-side validation for:
 - Date constraints (travel & expenses)
 - Required fields
 - File types and sizes
- Client-side validation for better user experience

Security Practices

- JWT-based authentication
- Role-based authorization checks
- Protection against SQL Injection and XSS attacks
- Secure file handling in cloud storage

Documentation

- API documentation using Postman collections
- Module-level documentation for workflows

Testing

- API testing using Postman
- Manual testing of UI workflows

Performance Optimization

- Optimized SQL queries
- Pagination for large datasets (feeds, expenses)
- Efficient state management using React Query
- Lazy loading of components

6.1 Testing Approach

The HRMS system was tested using manual testing techniques to ensure all modules function correctly according to requirements. Test cases were designed to validate:

- Input field validations
- Business rules (date constraints, limits, dependencies)
- File upload restrictions
- Role-based behavior
- Successful form submissions and error handling

6.2 Test Cases and Results

- Add Travel Form

[Table 6.2.1 Test Case for Travel Form]

TC ID	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status
T1	Travel Title	Valid input	Business Trip	Accepted	Accepted	Pass
T2	Travel Title	Empty input	—	Validation Error	Validation Error	Pass
T3	Travel Title	Exceeds max length	300 chars	Validation Error	Validation Error	Pass
T4	Description	Valid input	Client meeting	Accepted	Accepted	Pass
T5	Description	Empty input	—	Validation Error	Validation Error	Pass
T6	Location	Valid input	Delhi	Accepted	Accepted	Pass
T7	Location	Empty input	—	Validation Error	Validation Error	Pass
T8	Start Date	Valid date	10-06-2026	Accepted	Accepted	Pass
T9	Start Date	Empty	—	Validation Error	Validation Error	Pass
T10	End Date	Valid date	15-06-2026	Accepted	Accepted	Pass
T11	End Date	Before start date	Invalid range	Validation Error	Validation Error	Pass
T12	End Date	Empty	—	Validation Error	Validation Error	Pass
T13	Max Amount	Valid	2000	Accepted	Accepted	Pass

T14	Max Amount	Empty	—	Validation Error	Validation Error	Pass
T15	Max Amount	Negative	-500	Validation Error	Validation Error	Pass
T16	Max Amount	Non-numeric	abc	Validation Error	Validation Error	Pass
T17	Submit	Valid form	Valid data	Record created	Record created	Pass
T18	Cancel	Cancel action	—	Redirect to list	Redirected	Pass

- **Job Opening Create Form**

[Table 6.2.2 Test Case for Job Opening]

TC ID	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status
J1	Title	Valid	Software Engineer	Accepted	Accepted	Pass
J2	Title	Empty	—	Validation Error	Validation Error	Pass
J3	Title	Too long	300 chars	Validation Error	Validation Error	Pass
J4	Job Role	Valid	Backend Dev	Accepted	Accepted	Pass
J5	Job Role	Empty	—	Validation Error	Validation Error	Pass
J6	Place	Valid	Ahmedabad	Accepted	Accepted	Pass
J7	Place	Empty	—	Validation Error	Validation Error	Pass
J8	Requirements	Valid	Java, SQL	Accepted	Accepted	Pass
J9	Requirements	Empty	—	Validation Error	Validation Error	Pass
J10	JD Upload	Valid PDF	JD.pdf	Accepted	Accepted	Pass
J11	JD Upload	Invalid file	JD.png	Validation Error	Validation Error	Pass
J12	JD Upload	Empty	—	Validation Error	Validation Error	Pass
J13	HR Select	Valid	HR selected	Added successfully	Added	Pass
J14	HR Select	Multiple	2 HRs	Added	Added	Pass
J15	HR Select	None	—	Validation Error	Validation Error	Pass
J16	Reviewer	Valid	Reviewer selected	Added	Added	Pass
J17	Reviewer	Multiple	2 users	Added	Added	Pass
J18	Reviewer	None	—	Validation Error	Validation Error	Pass

J19	Submit	Valid form	Valid data	Job created	Job created	Pass
J20	Submit	Missing fields	No title	Validation Error	Validation Error	Pass

- **Game Create Form**

[Table 6.2.3 Test Case for Game Module]

TC ID	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status
G1	Game Name	Valid	Chess	Accepted	Accepted	Pass
G2	Game Name	Empty	—	Validation Error	Validation Error	Pass
G3	Game Name	Too long	300 chars	Validation Error	Validation Error	Pass
G4	Max Player	Valid	4	Accepted	Accepted	Pass
G5	Max Player	Negative	-2	Validation Error	Validation Error	Pass
G6	Max Player	Non-numeric	abc	Validation Error	Validation Error	Pass
G7	Min Player	Valid	2	Accepted	Accepted	Pass
G8	Min Player	> Max	5 > 4	Validation Error	Validation Error	Pass
G9	Duration	Valid	30	Accepted	Accepted	Pass
G10	Duration	Negative	-10	Validation Error	Validation Error	Pass
G11	Slot Time	Valid	60	Accepted	Accepted	Pass
G12	Slot Days	Valid	7	Accepted	Accepted	Pass
G13	Submit	Valid form	Valid data	Game created	Game created	Pass

- **Post Create Form**

[Table 6.2.4 Test Case for Post Creat]

TC ID	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status
P1	Title	Valid	Team Achievement	Accepted	Accepted	Pass
P2	Title	Empty	—	Validation Error	Validation Error	Pass
P3	Description	Valid	Project completed	Accepted	Accepted	Pass
P4	Description	Empty	—	Validation Error	Validation Error	Pass
P5	Image	Valid	post.jpg	Accepted	Accepted	Pass

P6	Image	Invalid	post.pdf	Validation Error	Validation Error	Pass
P7	Tags	Valid	celebration	Added	Added	Pass
P8	Tags	Multiple	success, team	Added	Added	Pass
P9	Visibility	Public	True	Public post	Public	Pass
P10	Submit	Valid form	Valid data	Post created	Post created	Pass
P11	Submit	Missing fields	No title	Validation Error	Validation Error	Pass

- **User Create Form**

[Table 6.2.5 Test Case for User]

TC ID	Field Name	Test Scenario	Input Data	Expected Result	Status
U1	Full Name	Valid	Joe Due	Accepted	Pass
U2	Full Name	Empty	—	Validation Error	Pass
U3	Email	Valid	<u>user@gmail.com</u>	Accepted	Pass
U4	Email	Invalid	usergmail.com	Validation Error	Pass
U5	Email	Duplicate	<u>hr@gmail.com</u>	Server Error	Pass
U6	Password	Valid	User@123	Accepted	Pass
U7	Password	Weak	12345	Validation Error	Pass
U8	Role	Selected	Employee	Accepted	Pass
U9	Department	Selected	IT	Accepted	Pass
U10	DOB & DOJ	Invalid relation	DOJ < DOB	Validation Error	Pass
U11	Submit	Valid form	Valid data	User created	Pass

- **Expense Create Form**

[Table 6.2.6 Test Case for Expense Create]

TC ID	Field Name	Test Scenario	Input Data	Expected Result	Status
E1	Details	Valid	Taxi fare	Accepted	Pass
E2	Details	Empty	—	Validation Error	Pass
E3	Amount	Valid	1500	Accepted	Pass
E4	Amount	Negative	-100	Validation Error	Pass
E5	Date	Valid	10-06-2026	Accepted	Pass
E6	Date	خرج travel	Invalid	Validation Error	Pass
E7	Category	Selected	Travel	Accepted	Pass
E8	Proof	Valid PDF	receipt.pdf	Accepted	Pass
E9	Proof	Invalid	.exe	Validation Error	Pass
E10	Submit	Valid form	Valid data	Submitted	Pass

Chapter 7

Conclusion and Future Extensions

7.1 Conclusion

The **Human Resource Management System (HRMS)** project successfully delivers a comprehensive and efficient web-based platform for managing core organizational processes. The system integrates multiple modules such as Travel & Expense Management, Social Engagement, Game Scheduling, Organization Chart, Recruitment, and Master Data Configuration into a unified solution.

The application enables streamlined workflows for travel planning, expense submission and approval, employee engagement, and recruitment processes. The implementation of role-based access control ensures that Employees, Managers, and HR/Admin users can securely perform only authorized actions within the system.

Modern technologies such as **React (Vite + TypeScript)** for the frontend and **Spring Boot** for the backend have been effectively utilized to build a scalable and maintainable architecture. The use of **Azure MySQL Database** and **Azure Blob Storage** ensures secure and reliable data management, especially for handling documents like receipts, travel files, and CV uploads.

Key features such as:

- Automated notifications (email and in-app)
- Validation rules for business logic (expense limits, deadlines)
- Fair scheduling algorithm for games
- Social engagement features with moderation
- Audit logging and secure authentication

enhance the overall usability, transparency, and efficiency of the system.

Through structured development, testing, and implementation, the HRMS system achieves its objective of reducing manual effort, improving data accuracy, and increasing productivity within an organization. The system is user-friendly, reliable, and adaptable to organizational needs, making it a strong foundation for future enhancements.

7.2 Future Enhancements

Although the current HRMS system fulfills the primary requirements, several enhancements can be implemented to further improve its capabilities and performance.

1. Advanced Analytics and Reporting

- Implementation of dashboards for HR insights such as:
 - Employee performance trends
 - Expense analytics
 - Travel cost summaries
- Graphical reports for better decision-making

2. AI-Based Features

- Smart recommendations for:
 - Expense anomaly detection
 - Candidate shortlisting in recruitment
- Predictive analytics for employee engagement and retention

3. Workflow Automation

- Automated approval chains for travel and expenses
- Configurable workflows based on department or role
- Reminder notifications for pending approvals

4. Integration with External Systems

- Integration with payroll systems
- Third-party calendar integration (e.g., Google Calendar, Outlook)
- Email service enhancements for better tracking

5. Security Enhancements

- Multi-factor authentication (MFA)
- Advanced audit logs and monitoring
- Data encryption for sensitive information

6. Scalability Improvements

- Microservices-based architecture for large-scale deployment
- Load balancing and performance optimization
- Support for multi-organization (multi-tenant architecture)

Chapter 8

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